A Family of Information-theoretic de Finetti Theorems for Constrained Optimization

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Polynomial Optimization

The setting: polynomial optimization

Many quantum and GPT problems take the form

$$\begin{aligned} p^* = \inf_{(x_A,x_B)\in K_A\times K_B} \quad & p(x_A,x_B) \\ \text{s.t.} \quad & f(x_A,x_B) = 0, \\ & g(x_A,x_B) \geq 0, \end{aligned} \tag{1}$$

where K_A, K_B are **convex bodies** (state spaces, compact, convex and nonempty subset of finite dimensional real vector space) and p, f, g are affine.

- includes quantum separability, non-local game values, quantum error correction and many more (GPT) problems [DPS04; Ber+21; PLG25; Tav+24]
- generally intractable [Gha10; Ioa07]

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- generally intractable [Gha10; Ioa07]

Goal: an outer and inner approximation scheme with quantitative convergence guarantees, compatible with equality and inequality constraints and numerically implementable

From affine to linear: tensor product reformulation

For affine $f : K_A \times K_B \to V$ there is a unique linear $\widehat{F}_{\hookrightarrow} : K_A \dot{\otimes} K_B \to V$.

We obtain the equivalent problem

$$\begin{aligned} p^* = \inf_{x_A \otimes x_B \in K_A \dot{\otimes} K_B} \quad & \widehat{P}_{\hookrightarrow}(x_A \otimes x_B) \\ \text{s. t.} \quad & \widehat{F}_{\hookrightarrow}(x_A \otimes x_B) = 0, \\ & \widehat{G}_{\hookrightarrow}(x_A \otimes x_B) \geq 0. \end{aligned} \tag{2}$$

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Two natural tensor products [Aub+21; Plá23]:

- **minimal**: $K_A \dot{\otimes} K_B = \mathrm{conv}\{x_A \otimes x_B : x_A \in K_A, x_B \in K_B\}$ (**separable states**)
- **maximal**: $K_A \hat{\otimes} K_B$ (**all entangled states**)
- $K_A \dot{\otimes} K_B \subseteq K_A \hat{\otimes} K_B$

$$C_1 \hat{\otimes} C_2 := (C_A^\star \dot{\otimes} C_B^\star)^\star$$
$$= \{z \in V_1 \otimes V_2 \ : \ (f_1 \otimes f_2)(z) \geq 0 \text{ for all } f_1 \in C_1^\star,\ f_2 \in C_2^\star\}.$$

Four flavours of the optimization problem

p_1: products in $K_A \dot{\otimes} K_B$

$$\begin{aligned} \inf\ & \widehat{P}_{\hookrightarrow}(x_A \otimes x_B) \\ \text{s.t. } & x_A \otimes x_B \in K_A \dot{\otimes} K_B \\ & \widehat{F}_{\hookrightarrow}(x_A \otimes x_B) = 0,\ \widehat{G}_{\hookrightarrow}(x_A \otimes x_B) \geq 0 \end{aligned}$$

$p_2 = p_1$: products in $K_A \hat{\otimes} K_B$

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p_3: separable in $K_A \dot{\otimes} K_B$

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p_4: all of $K_A \hat{\otimes} K_B$

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$$p_4 \leq p_3 \leq p_1 = p_2$$

The gap $p_1 \neq p_3$ is the subtle point: constraints need not pass to extreme points, but p_1 **non-convex**

Convexification of p_1

Convexity: Two equivalent problems:

p_1: products in $K_A \dot{\otimes} K_B$

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$\tilde{p}_1$: "for every λ"

$$\begin{aligned} &\inf \widehat{P}_{\hookrightarrow}(x_{AB}) \\ &\text{s.t. } x_{AB} = \sum_{\lambda} p(\lambda) x_A^{\lambda} \otimes x_B^{\lambda} \in K_A \dot{\otimes} K_B \\ &\quad \forall \lambda : \; \widehat{F}_{\hookrightarrow}(x_A^{\lambda} \otimes x_B^{\lambda}) = 0, \; \widehat{G}_{\hookrightarrow}(x_A^{\lambda} \otimes x_B^{\lambda}) \geq 0 \end{aligned}$$

$$p_3 \leq p_1 = \tilde{p}_1$$

DPS-type hierarchy: symmetric extensions

A state $x_{AB} \in K_A \hat{\otimes} K_B$ is ***n*-extendible** if it has a permutation-invariant extension on $K_A \hat{\otimes} K_B^{\hat{\otimes} n}$ [DPS04; BL09].

- Denote the set by $\mathrm{Ext}_n(K_A, K_B)$.
- $\mathrm{Ext}_{n+1}(K_A : K_B) \subseteq \mathrm{Ext}_n(K_A : K_B)$, and by [AMP24]

$$\bigcap_{n \geq 1} \mathrm{Ext}_n(K_A, K_B) = K_A \dot{\otimes} K_B. \tag{3}$$

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$\Rightarrow$ DPS hierarchy converges to p_3 **asymptotically**.

But: convergence holds only **on average**.

- $x_{AB} = \sum_\lambda p(\lambda) x_A^\lambda \otimes x_B^\lambda$
- $\widehat{F}_{\hookrightarrow}(x_{AB}) = 0$ does not imply $\widehat{F}_{\hookrightarrow}(x_A^\lambda \otimes x_B^\lambda) = 0$ for every λ.
- No finite-level error bounds.
- Inequality constraints are not handled.

The polarization hierarchy [PLG25]

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Idea: symmetric **double** extensions $\left(K_A \hat{\otimes} K_B\right)^{\hat{\otimes} n}$, plus a **polarization** constraint enforcing equality on extreme points [PLG25; LG23].

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- Convergence to p_1 for **global equality** constraints.
- Resolves the on-average issue of DPS.

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- No **inequality** constraints.
- Heuristic interior points only.

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This work: a third route via information-theoretic de Finetti theorems.

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Comparison at a glance

We restrict the constraint structure to the **local** form

$$f(x_A, x_B) = f_A(x_A) \oplus f_B(x_B), \tag{4}$$

i.e. $\widehat{F}_{\hookrightarrow,A}(x_A) = 0$, $\widehat{F}_{\hookrightarrow,B}(x_B) = 0$, $\widehat{G}_{\hookrightarrow,A}(x_A) \geq 0$, $\widehat{G}_{\hookrightarrow,B}(x_B) \geq 0$.

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Goal: Extending the paradigm of symmetric extensions with improved convergence guarantees to this setting

	Polarization [PLG25]	This work
Constraints	global equality	local eq. & ineq.
Conv. rate	asymptotic	quantitative finite-n
Interior points	heuristic	certified, with bound

Relative entropy and mutual information for convex bodies

The first obstacle

In quantum theory we use the Umegaki relative entropy [Ume62]

$$D(\rho\|\sigma) = \mathrm{tr}[\rho(\log\rho - \log\sigma)]. \tag{5}$$

In GPTs:

- No canonical entropy [KNI10; KIF16; Tak19] – GPTs typically lack a logarithm.
- But: we need
  - Pinsker-type inequality $D(\rho\|\sigma) \geq \frac{1}{2}\|\rho - \sigma\|_1^2$
  - positivity $D(\rho\|\sigma) \geq 0$
  - Data-Processing inequality $D(\rho\|\sigma) \geq D(\Phi(\rho)\|\Phi(\sigma))$, Φ channel

Solution: a recently proposed integral representation of the quantum relative entropy [Fre23] extends naturally to GPTs.

Integral representation of relative entropy

For $x, y \in K$, following [Fre23],

$$D(x\|y) := \int_{\mathbb{R}} \frac{dt}{|t|(t-1)^2} \sup_{f \in E(K)} -\big[(1-t)f(x) + tf(y)\big]. \qquad (6)$$

Properties (cf. [Fre23; Jen24]):

- DPI under all positive maps,
- homogeneity $D(\lambda x\|\lambda y) = \lambda D(x\|y)$,
- coincides with D_{KL} on simplices.

Q: mutual information $I(A:B)_\rho = D(\rho_{AB}\|\rho_A \otimes \rho_B)$

Variational form of mutual information (cf. [Wil16]):

$$I(A:B)_{x_{AB}} := \inf_{y_A \in K_A} D(x_{AB}\|y_A \otimes x_B), \quad x_{AB} \in K_A \hat{\otimes} K_B. \qquad (7)$$

Useful technical tools

Throughout we use two tools repeatedly.

Pinsker-type inequality

For all $x, y \in K$ and $\lambda \in \mathbb{R}_+$,

$$\frac{1}{2}\left(\sup_{f \in E(K)} f(x) - f(y)\right)^2 \leq \int_0^\lambda \frac{ds}{s}\left[\sup_{f \in E(K)} sf(y) - f(x)\right] + \ln\lambda + 1 - \lambda. \tag{8}$$

Logarithmic upper bound

If $x \leq \lambda y$ in the cone $A(K)^{\star +}$, then

$$D(x\|y) \leq 1 + \ln\lambda. \tag{9}$$

Proofs adapt [Fre23; CK11] to the conic setting.

Monogamy of mutual information

The second obstacle: is $I(A:B)$ bounded?

In quantum theory: $I(A:B) \leq 2\log d_A$ [Wil16].

In GPTs, we ask:

$$I(A:B)_{x_{AB}} \overset{?}{\leq} c(A), \quad x_{AB} \in K_A \hat{\otimes} K_B, \tag{10}$$

with $c(A)$ depending only on K_A, **not on** K_B.

Key geometric input: a uniform interior reference state, building on results of [AMP24].

The uniform reference

Uniform order unit bound

Let K be a state space and $\tau \in \mathrm{relint}(K)$. Then

$$\lambda := \sup_{x \in K} \inf\{\mu \in \mathbb{R}_+ : x \leq \mu\tau\} < \infty. \tag{11}$$

Interpretation:

- Any state τ in the **relative interior** of K acts as an order unit [Roc97; HW20].
- All other states are uniformly dominated: $x \leq \lambda\tau$ for every $x \in K$.
- The constant $\lambda = \lambda(K, \tau)$ encodes the geometry of K.

$\Rightarrow$ we have a uniform reference to compare against.

The conic mutual information bound

Conic mutual information bound

For $x_{AB} \in K_A \hat{\otimes} K_B$,

$$I(A:B)_{x_{AB}} \;\leq\; \lambda_A\big(1 + \ln \lambda_A\big), \tag{12}$$

where λ_A is the constant from the previous lemma applied to K_A.

- Entanglement in GPTs is inherently **monogamous** [AMP24].
- This generalizes the quantum monogamy estimate to arbitrary convex bodies.
- The constant $c(A)$ depends **only** on the geometry of K_A.

A finite de Finetti theorem for convex bodies

The main theorem

Theorem (Finite de Finetti for convex bodies)

If $x_{AB} \in K_A \hat{\otimes} K_B$ is permutation invariant n-extendible, then there exists $\widetilde{x}_{AB} \in K_A \dot{\otimes} K_B$ such that

$$\left\| \widetilde{x}_{AB} - x_{AB} \right\|_{\star, 1_{K_A \hat{\otimes} K_B}} \leq \frac{2\, c(A,B)}{\sqrt{n}}, \tag{13}$$

where $c(A,B)$ depends only on K_A and K_B.

Reading the statement:

- maximal-tensor-product states with permutation symmetry are close to minimal-tensor-product states,
- at a quantitative rate $1/\sqrt{n}$,
- with the constant given **explicitly** in terms of K_A, K_B.
- **Quantum:** Improvement to $\mathcal{O}(1/n)$ via group-theoretic arguments and double-sided de Finetti
- **Quantum:** Improvement of $c(A,B)$ via group-theoretic arguments and fixed point de Finetti

Compare with the qualitative result of [AMP24; BL09].

Proof sketch I: Mutual information bound on simplices

Let $y_{AB_1^n} \in K_A \hat{\otimes} K_B^{\hat{\otimes} n}$ be the permutation-invariant extension of x_{AB}.

Step 1. Pick informationally complete (injective maps) measurements $\mathcal{M}_{A \to Y}$, $\mathcal{M}_{B \to Z}$.

Step 2. Set

$$y_{AZ^n} := \left(\mathrm{id}_A \otimes \mathcal{M}_{B \to Z}^{\otimes n}\right)\left(y_{AB_1^n}\right), \quad y_{YZ^n} := \left(\mathcal{M}_{A \to Y} \otimes \mathcal{M}_{B \to Z}^{\otimes n}\right)\left(y_{AB_1^n}\right). \tag{14}$$

Step 2. Apply conic mutual information bound:

$$I(A : Z_1^n)_{y_{AZ_1^n}} \leq c(A). \tag{15}$$

By DPI [Fre23],

$$I(Y : Z_1^n)_{y_{YZ_1^n}} \leq c(A). \tag{16}$$

Proof sketch II: Monogamy bound

Step 4. The simplex side is classical: the chain rule applies [CK11; Wil16],

$$I(Y:Z_1^n) = \sum_{m=0}^{n-1} \sum_{z^m} p(z^m)\, I(Y:Z_{m+1})_{y_{YZ_{m+1}|z^m}}. \quad (17)$$

Step 5. By positivity and pigeonhole, there exists $m \in \{0, \ldots, n-1\}$ with

$$\sum_{z^m} p(z^m)\, I(Y:Z_{m+1})_{y_{YZ_{m+1}|z^m}} \;\le\; \frac{c(A)}{n}. \quad (18)$$

Pinsker on the simplex side together with convexity gives

$$\sup_{f \in E(K_{YZ_{m+1}})} f\Big(y_{YZ_{m+1}} - \sum_{z^m} p(z^m)\, y_{Y|z^m} \otimes y_{Z_{m+1}|z^m}\Big) \;\le\; \sqrt{\frac{2c(A)}{n}}. \quad (19)$$

Proof sketch 3: lift back via informational completeness

Step 6. For IC measurements, there exist constants $\tilde{f}(A, B, \mathcal{M}_{A\to Y}) > 0$ and $\tilde{f}(A, B, \mathcal{M}_{B\to Z}) > 0$ such that, for every $x' \in K_A \hat{\otimes} K_B$ we have

$$\sup_{g\in E(K_{YZ})} g\left(\left(\mathcal{M}_{A\to Y} \otimes \mathcal{M}_{B\to Z}\right)(x')\right) \geq \underbrace{\tilde{f}(A, B, \mathcal{M}_{A\to Y})\tilde{f}(A, B, \mathcal{M}_{B\to Z})}_{=:f(A,B)} \sup_{h\in E(K_A\hat{\otimes}K_B)} h(x'). \tag{20}$$

With $x' = y_{AB_{m+1}} - \sum_{z^m} p(z^m) y_{A|z^m} \otimes y_{B_{m+1}|z^m}$ we obtain

$$\sup_{h\in E(K_A\hat{\otimes}K_{B_{m+1}})} h\left(y_{AB_{m+1}} - \sum_{z^m} p(z_1^m) y_{A|z_1^m} \otimes y_{B_{m+1}|z^m}\right) \leq \frac{\sqrt{2c(A)}}{\sqrt{n} f(A, B)}. \tag{21}$$

Step 7. Permutation invariance and a complement-effect argument [Plá23] turns this into the dual order-unit norm.

A convergent hierarchy for polynomial optimization

From the finite de Finetti theorem to a hierarchy

Recap. For any permutation-symmetric $x_{AB_1^n} \in K_A \hat{\otimes} K_B^{\hat{\otimes} n}$, there is a separable state $\tilde{x}_{AB} \in K_A \dot{\otimes} K_B$ with

$$\|\tilde{x}_{AB} - x_{AB}\| \leq \frac{c(A,B)}{\sqrt{n}}, \tag{22}$$

where $c(A,B)$ depends only on K_A, K_B [Zei+26].

From the finite de Finetti theorem to a hierarchy

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Strategy. Relax the intractable (non-linear) problem over $K_A \dot{\otimes} K_B$ to one over the **symmetric extensions**:

- enforce the local constraints on one B-extension
- Symmetry of states translates to symmetry of constraint maps
- By construction: Constraint maps commute with IC-measurement in de Finetti proof
- use the de Finetti bound to translate this into an explicit error in $\sqrt{n}^{-1}$.

The hierarchy

For $n \in \mathbb{N}$, define the level-n outer relaxation

$$
\begin{aligned}
p^{(n)} \;=\; \inf_{x_{AB_1^n}} \quad & \widehat{P}_{\hookrightarrow}(x_{AB_1}) \\
\text{s.t.} \quad & x_{AB_1^n} \in K_A \hat{\otimes} K_B^{\hat{\otimes} n}, \\
& x_{AB_1^n} \text{ permutation-invariant on } B_1, \ldots, B_n, \\
& \widehat{F}_{A,\hookrightarrow} \otimes \mathrm{id}_{B^n}(x_{AB_1^n}) = 0, \quad \mathrm{id}_A \otimes \widehat{F}_{B,\hookrightarrow} \otimes \mathrm{id}_{B^{n-1}}(x_{AB_1^n}) = 0, \\
& \widehat{G}_{A,\hookrightarrow} \otimes \mathrm{id}_{B^n}(x_{AB_1^n}) \geq 0, \quad \mathrm{id}_A \otimes \widehat{G}_{B,\hookrightarrow} \otimes \mathrm{id}_{B^{n-1}}(x_{AB_1^n}) \geq 0.
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\end{aligned}
$$

Outer hierarchy:

$$
p^{(1)} \leq p^{(2)} \leq \ldots \leq p^{(n)} \leq \ldots \leq p_3. \tag{23}
$$

Key difference to DPS [DPS04]: the local constraints are enforced **on every marginal**, not only on average.

Convergence with explicit finite-n rate

Theorem (Convergence of the hierarchy)

Under mild regularity assumptions on $K_A, K_B, p, f_A, f_B, g_A, g_B$,

$$\boxed{0 \;\leq\; p_3 - p^{(n)} \;\leq\; \frac{2\left\|\widehat{P}_{\hookrightarrow}\right\|_{1_{K_A \hat{\otimes} K_B}} \cdot c(A,B)}{\sqrt{n}}} \tag{24}$$

where $c(A,B)$ is the de Finetti constant depending on K_A, K_B.

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- First quantitative finite-level guarantee for general convex bodies.
- Handles equality and inequality constraints simultaneously.
- Additional group-theoretic arguments recovers DPS/polarization rates in the quantum case [BH17; Chr+07].

Generating inner points

The level-n outer solution $x^{(n)}_{AB_1^n}$ is not yet a feasible product state.

Rounding scheme (cf. [Koß+25; Zei+25]):

1. Take the de Finetti candidate $\tilde{x}_{AB} = \sum_\lambda p_\lambda \, x_A^\lambda \otimes x_B^\lambda$ obtained from $x^{(n)}_{AB_1^n}$ by measurement rounding
2. Among the components $\{x_A^\lambda \otimes x_B^\lambda\}_\lambda$, pick the one minimizing the objective subject to the local constraints.

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The level-n outer solution $x^{(n)}_{AB_1^n}$ is not yet a feasible product state.

Rounding scheme (cf. [Koß+25; Zei+25]):

1. Take the de Finetti candidate $\tilde{x}_{AB} = \sum_\lambda p_\lambda \, x_A^\lambda \otimes x_B^\lambda$ obtained from $x^{(n)}_{AB_1^n}$ by measurement rounding
2. Among the components $\{x_A^\lambda \otimes x_B^\lambda\}_\lambda$, pick the one minimizing the objective subject to the local constraints.

Output: a certified **feasible** interior point $x_A^\star \otimes x_B^\star \in K_A \dot{\otimes} K_B$.

Conceptually: the de Finetti theorem provides a **constructive bridge** from the outer relaxation to a genuine feasible point.

Sandwich bound on the original value

Theorem (Convergent inner sequence [Koß+25; Zei+25])

Let $p_{\mathrm{in}}^{(n)}$ be the objective value at the rounded interior point. Then

$$p^{(n)} \le p_3 \le p_{\mathrm{in}}^{(n)}, \qquad p_{\mathrm{in}}^{(n)} - p^{(n)} \le \mathcal{O}\left(\frac{1}{\sqrt{n}}\right). \tag{25}$$

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Consequence – a true sandwich:

$$p^{(n)} \text{ (outer, SDP/conic)} \longrightarrow \;\leq\; p_3 \;\leq\; \longleftarrow \; p_{\mathrm{in}}^{(n)} \text{ (inner, certified)}$$

Both bounds tighten at the rate $1/\sqrt{n}$. We obtain matching upper and lower bounds on the true optimum.

Representability: when can we compute the hierarchy?

The hierarchy is a **convex** optimization problem over $K_A \hat{\otimes} K_B^{\hat{\otimes} n}$.

Question. When is it numerically solvable?

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Examples:

- Quantum state spaces → SDP [DPS04; Wil16].
- Classical simplices → LP.
- Homogeneous self-dual cones → symmetric cone programming [GIL25; HN08; GPT13; Tho19].

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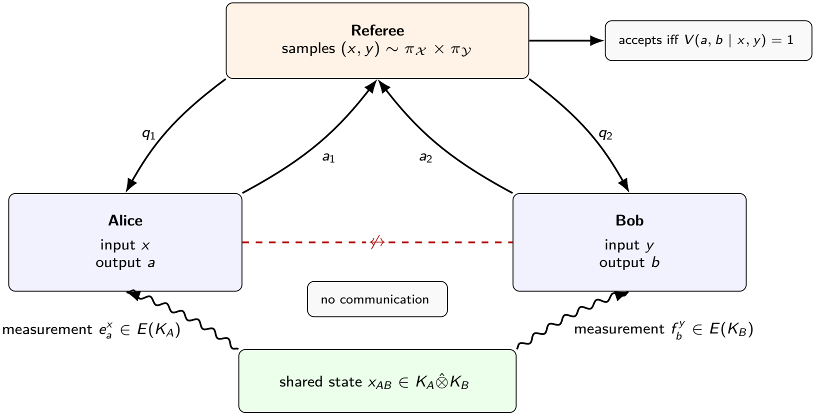
$\Rightarrow$ for these state spaces the entire scheme is numerically implementable.

Application: non-local games

Two-player non-local Games



GPT value of the game over state spaces K_A, K_B:

$$\omega_{\mathrm{GPT}} = \sup_{\omega_{AB}, \{e_a^x\}, \{f_b^y\}} \sum_{x,y,a,b} \pi_{\mathcal{X}}(x)\pi_{\mathcal{Y}}(y)V(a, b \mid x, y)\,\langle e_a^x \otimes f_b^y, \omega_{AB}\rangle,$$

Approximation scheme for non-local games

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Applying the approximation hierarchy yields:

Theorem (GPT-value approximation)

For any two-player non-local game over state spaces K_A, K_B admitting conic lifts, there is a sequence of conic programs producing

$$\omega_{\mathrm{GPT}}^{(n)} \to \omega_{\mathrm{GPT}} \qquad \textit{with rate } \mathcal{O}(1/\sqrt{n}), \tag{26}$$

*together with certified **feasible** strategies achieving values* $\omega_{\mathrm{GPT}} - \mathcal{O}(1/\sqrt{n})$.

Computational complexity

In particular (cf. [Jee+21; Zei+25; Ber+21]):

- Equality and inequality constraints (e.g. no-signalling, dimension,...) are handled natively and are compatible with our de Finetti theorems and group-theoretic symmetry reduction techniques.
- For quantum systems of bounded dimension: symmetry-based approximation algorithms with poly(ϵ^{-1}) performance based on Bose-symmetric de Finetti Theorem.
- For weakly self-dual GPTs: explicit rates in terms of geometric data of K_A, K_B.

Theorem

For a two-player free non-local game with $|A|$ answers, $|Q|$ questions and quantum assistance of size $|T|$, there exists a poly$\left(\epsilon^{-1}\right)$ costly transformation (see **??** *and* **??** *for further details) to an SDP with at most*

$$\left(\epsilon^{-4} \cdot \mathrm{poly}\left(|A|, |Q|, |T|\right)\right)^{(|A||Q||T|)^2}, \tag{27}$$

real degrees of freedom, which approximates the value of the game up to an additive ϵ-error.

Conclusion

What we did.

- Built an information-theoretic finite de Finetti theorem on convex bodies.
- Used it to obtain a convergent outer hierarchy with explicit $1/\sqrt{n}$ rate.
- Quantum: Improved convergence performance via group-theoretic arguments and several novel de Finetti theorems
- Constructed certified inner points by measurement rounding to the de Finetti candidate.
- Sandwiched the optimum of polynomial optimization problems and recast non-local games inside the framework.

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Open directions.

- More general global constraints [PLG25].
- Other applications (see e.g. approximate quantum error correction [Koß+25] and channel discrimination [Ohs+24]).

Thank you!

arXiv:2601.15184 , arXiv:2507.12302 , arXiv:2507.12326

References I

[Aub+21] Guillaume Aubrun et al. "Entangleability of cones". In: **Geometric and Functional Analysis** 31.2 (Apr. 2021), pp. 181–205.

[AMP24] Guillaume Aubrun et al. "Monogamy of entanglement between cones". In: **Mathematische Annalen** 391.1 (Aug. 2024), pp. 1591–1609.

[BL09] Jonathan Barrett and Matthew Leifer. "The de Finetti theorem for test spaces". In: **New Journal of Physics** 11.3 (Mar. 2009), p. 033024.

[Ber+21] Mario Berta et al. "Semidefinite programming hierarchies for constrained bilinear optimization". In: **Mathematical Programming** 194.1–2 (Apr. 2021), pp. 781–829.

[BH17] Fernando G. S. L. Brandão and Aram W. Harrow. "Quantum de Finetti Theorems Under Local Measurements with Applications". In: **Communications in Mathematical Physics** 353.2 (Apr. 2017), pp. 469–506.

[Chr+07] Matthias Christandl et al. "One-and-a-Half Quantum de Finetti Theorems". In: **Communications in Mathematical Physics** 273.2 (Mar. 2007), pp. 473–498.

[CK11] Imre Csiszár and János Körner. **Information Theory: Coding Theorems for Discrete Memoryless Systems**. Cambridge University Press, June 2011.

References II

[DPS04] Andrew C. Doherty et al. "Complete family of separability criteria". In: **Physical Review A** 69.2 (Feb. 2004).

[Fre23] Péter E. Frenkel. "Integral formula for quantum relative entropy implies data processing inequality". In: **Quantum** 7 (Sept. 2023), p. 1102.

[Gha10] S. Gharibian. "Strong NP-hardness of the quantum separability problem". In: **Quantum Information and Computation** 10.3 and 4 (Mar. 2010), pp. 343–360.

[GPT13] João Gouveia et al. "Lifts of Convex Sets and Cone Factorizations". In: **Mathematics of Operations Research** 38.2 (May 2013), pp. 248–264.

[GIL25] João Gouveia et al. "Faces of homogeneous cones and applications to homogeneous chordality". 2025.

[HN08] J. William Helton and Jiawang Nie. "Semidefinite representation of convex sets". In: **Mathematical Programming** 122.1 (Aug. 2008), pp. 21–64.

[HW20] Daniel Hug and Wolfgang Weil. **Lectures on Convex Geometry**. Springer International Publishing, 2020.

[Ioa07] L.M. Ioannou. "Computational complexity of the quantum separability problem". In: **Quantum Information and Computation** 7.4 (May 2007), pp. 336–370.

References III

[Jee+21] Hyejung H. Jee et al. "Quasi-Polynomial Time Algorithms for Free Quantum Games in Bounded Dimension". en. In: **LIPIcs, Volume 198, ICALP 2021** 198 (2021), 82:1–82:20.

[Jen24] Anna Jenčová. "Recoverability of quantum channels via hypothesis testing". In: **Letters in Mathematical Physics** 114.1 (Feb. 2024).

[KNI10] Gen Kimura et al. "Distinguishability measures and entropies for general probabilistic theories". In: **Reports on Mathematical Physics** 66.2 (2010), pp. 175–206.

[KIF16] Gen Kimura et al. "Entropies in general probabilistic theories and their application to the Holevo bound". In: **Physical Review A** 94.4 (2016), p. 042113.

[Koß+25] Gereon Koßmann et al. **On approximate quantum error correction for symmetric noise**. 2025.

[LG23] Laurens T. Ligthart and David Gross. "The inflation hierarchy and the polarization hierarchy are complete for the quantum bilocal scenario". In: **Journal of Mathematical Physics** 64.7 (July 2023).

[Ohs+24] Ties-A. Ohst et al. **Characterising memory in quantum channel discrimination via constrained separability problems**. 2024.

References IV

[Plá23] Martin Plávala. "General probabilistic theories: An introduction". In: **Physics Reports** 1033 (Sept. 2023), pp. 1–64.

[PLG25] Martin Plávala et al. "The polarization hierarchy for polynomial optimization over convex bodies, with applications to nonnegative matrix rank". In: **Linear Algebra and its Applications** 723 (2025), pp. 15–32.

[Roc97] R.T. Rockafellar. **Convex Analysis**. Princeton Landmarks in Mathematics and Physics. Princeton University Press, 1997.

[Tak19] Ryo Takakura. "Entropy of mixing exists only for classical and quantum-like theories among the regular polygon theories". In: **Journal of Physics A: Mathematical and Theoretical** 52.46 (2019), p. 465302.

[Tav+24] Armin Tavakoli et al. "Semidefinite programming relaxations for quantum correlations". In: **Reviews of Modern Physics** 96.4 (Dec. 2024).

[Tho19] Rekha R. Thomas. "Spectrahedral Lifts Of Convex Sets". In: **Proceedings of the International Congress of Mathematicians (ICM 2018)**. WORLD SCIENTIFIC, May 2019, pp. 3819–3842.

[Ume62] Hisaharu Umegaki. "Conditional expectation in an operator algebra. IV. Entropy and information". In: **Kodai Mathematical Seminar Reports** 14.2 (1962), pp. 59–85.

References V

[Wil16] Mark M. Wilde. **Quantum Information Theory**. Cambridge University Press, Nov. 2016.

[Zei+25] Julius A. Zeiss et al. **Approximating fixed size quantum correlations in polynomial time**. 2025.

[Zei+26] Julius A. Zeiss et al. **Finite de Finetti for convex bodies and Polynomial Optimization**. 2026.

The connection to information theory

Let $\mathcal{M} : B \to Z$ be a measurement.

$$x_{AZ_1^n} := \mathrm{id}_A \otimes \mathcal{M}_{B\to Z}^{\otimes n}(x_{AB_1^n}) \tag{28}$$

The self-decoupling lemma [BH17] yields then

$$I(A : Z_1^n)_{x_{AZ_1^n}} = \sum_{m=0}^{n-1} \sum_{z_1^m} p(z_1^m) \underbrace{D\left(x_{AZ_{m+1|z_1^m}} \| x_{A|z_1^m} \otimes x_{Z_{m+1|z_1^m}}\right)}_{\text{"distance to a product state"}}. \tag{29}$$

$I(A : B)$ is the mutual information and $D(\cdot\|\cdot)$ the relative entropy.

A proof sketch [Ber+21]

1. Given $x_{AB_1^n}$, then

$$I(A:B_1^n)_{x_{AB_1^n}} \le 2\log d_A. \tag{30}$$

2. Apply the measurements $\mathcal{M}_{B\to Z}$ and the self-decoupling lemma

$$\begin{aligned}\frac{2\log d_A}{n} &\ge \sum_{z_1^m} p(z_1^m)\, D\Big(x_{AZ_{m+1|z_1^m}} \Big\| x_{A|z_1^m} \otimes x_{Z_{m+1|z_1^m}}\Big) \\ &\ge \frac{1}{2\ln 2}\Big\| x_{AZ_{m+1}} - \sum_{z_1^m} x_{A|z_1^m} \otimes x_{Z_{m+1|z_1^m}}\Big\|_1^2.\end{aligned} \tag{31}$$

3. Assume

$$\Big\| x_{AZ_{m+1}} - \sum_{z_1^m} x_{A|z_1^m} \otimes x_{Z_{m+1|z_1^m}}\Big\|_1 \ge c(\mathcal{M})\,\Big\| x_{AB_{m+1}} - \sum_{z_1^m} p(z_1^m) x_{A|z_1^m} \otimes x_{B_{m+1|z_1^m}}\Big\|_1. \tag{32}$$

Under compatibility with constraints:

$$\Rightarrow \quad \lim_{n\to\infty} \Sigma^n(A:B) = \Sigma(A:B) \tag{33}$$